

ACADEMIC INTELLIGENCE & ACCREDITATION SYSTEM

AcademiQ Platform

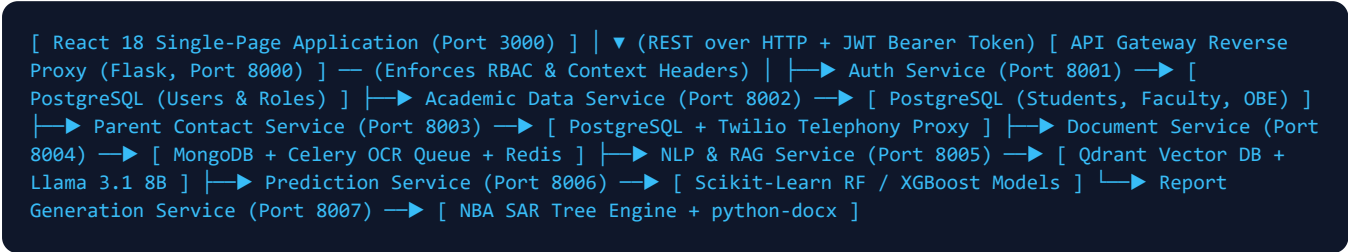
AI-Powered Unified Academic Intelligence, Accreditation Automation (NBA Tier-II GAPC V4.0), and Predictive Student Analytics

Executive Overview: AcademiQ is a comprehensive, production-ready microservices platform engineered to modernize higher education administration. It unifies outcome-based education (OBE) tracking, automated Self-Assessment Report (SAR) compilation for NBA accreditation, intelligent document OCR and RAG querying, real-time ML-driven student risk intervention, and DPDP Act-compliant privacy-first parent communications.

PROJECT TYPE	ACADEMIC BATCH
B.E. Computer Science & Engineering (Final Year Project)	Z10 Batch · Academic Year 2025–2026
ACCREDITATION STANDARD	CORE ARCHITECTURE
NBA UG Tier-II GAPC V4.0 (1000 Marks Framework)	8 Polyglot Microservices + 4 Specialized Databases
AI / ML ENGINE	PRIVACY & REGULATORY
Llama 3.1 8B (Groq/Ollama) + BGE-M3 + Random Forest & XGBoost	DPDP Act 2023 Compliant (Masked Identity + Twilio Proxy Bridge)

1. System Architecture & Topology

AcademiQ is designed as an event-driven, decoupled microservices ecosystem. A central API Gateway handles authentication, role-based access control (RBAC), context injection, and reverse-proxy routing to 7 backend domain microservices backed by 4 persistent database stores.



Infrastructure & Persistent Storage Layer

Data Store	Technology	Role & Managed Entities
Relational DB	PostgreSQL 16 Alpine	Users, roles, student academic records, faculty profiles, department curricula (PEOs, POs, COs), parent contact data, call logs, and report job statuses.
Document Store	MongoDB 7.0	Document metadata, parsed text chunks, ingestion status, OCR logs, and unstructured institution artifacts.
Vector Database	Qdrant v1.9.2	Dense 1024-dimensional vector embeddings generated by BAAI/bge-m3 for semantic search and RAG retrieval.
Message Broker & Cache	Redis 7 Alpine	Task queue backend for Celery asynchronous workers (OCR extraction, document embedding, background report rendering).

2. Microservices Directory & Responsibilities

Each service in AcademiQ adheres to single-responsibility design principles with dedicated storage boundaries and well-defined API contracts.

1. API Gateway

Port 8000

Responsibilities:

- Single entry point for all frontend client traffic.
- Validates JWT signatures, algorithm, and expiration.
- Fail-closed routing table with network-boundary RBAC.
- Injects user context headers (`X-User-Id` , `X-User-Role` , `X-Linked-Id`).

2. Auth Service

Port 8001

Responsibilities:

- User registration, login, and profile management.
- Bcrypt password hashing and secure token generation.
- Role assignments (`admin` , `teacher` , `student` , `worker`).
- Account linking to `student_id` or `faculty_id` .

3. Academic Data Service

Port 8002

Responsibilities:

- Manages department curricula, Vision/Mission, PEOs, POs, COs.
- Maintains 360-degree student profiles (8 academic metrics).
- Stores faculty qualifications, research, and FDP records.
- Supplies raw metrics to the prediction and report services.

4. Parent Contact Service

Port 8003

Responsibilities:

- Maintains parent contact registry and communication preferences.
- DPDP Act 2023 consent validation before communication.
- Phone number masking for faculty UI (`*****3210`).
- Twilio Proxy integration for bi-directional masked calls/SMS.

5. Document Service & Worker

Port 8004

Responsibilities:

- Multi-format ingestion: PDF, DOCX, PNG, JPG, TXT.
- Dual-engine OCR: PyMuPDF (digital) + PaddleOCR (scans).
- Sliding window chunking (512 tokens with 50 overlap).
- Celery task dispatch for background asynchronous processing.

6. NLP & RAG Service

Port 8005

Responsibilities:

- Vector embeddings via `BAAI/bge-m3` (1024-dim dense).
- Qdrant vector collection indexing and similarity retrieval.
- Context-grounded QA via Llama 3.1 8B (Groq / Ollama).
- Hallucination-free formal narrative generation for SAR.

7. Prediction Service

Port 8006

Responsibilities:

- Real-time academic risk prediction and failure classification.
- Primary Random Forest pipeline + secondary XGBoost model.
- At-risk cohort filtering and early warning threshold scoring.
- Automated continuous model retraining endpoint.

8. Report Generation Service

Port 8007

Responsibilities:

- NBA SAR tree compiler for UG Tier-II GAPC V4.0 (1000 marks).
- Autonomous formula engine (SFR, FQI, Cadre, API, Placement).
- LLM-driven narrative expansion from grounded factual bullets.
- Export pipeline for publication-ready DOCX and PDF documents.

3. Feature Deep-Dive: NBA SAR Accreditation Engine

AcademiQ features an implementation of the National Board of Accreditation (NBA) Self-Assessment Report (SAR) framework for **UG Tier-II Engineering (GAPC V4.0 — January 2025 Standard)**. The system models the complete 1000-mark evaluation tree across 9 root criteria, Part C declaration, and 3 annexures.

The 1000-Mark Evaluation Framework

Criterion	Official NBA Criteria Title	Max Marks	Data Source & Calculation Method
Criterion 1	Outcome-Based Curriculum (Vision, Mission, PEOs, PO/CO Mapping)	120 Marks	Department Curricula + Matrix Tables
Criterion 2	Teaching-Learning Processes (Pedagogy, Projects, SDGs, Internships)	120 Marks	Manual Records + AI Structured Narratives
Criterion 3	Assessment & CO/PO Attainment (Continuous Eval, Direct/Indirect)	120 Marks	Formula Engine: <code>co_attainment</code> , <code>po_attainment</code>
Criterion 4	Students' Performance (Enrolment, Success Rate, API, Placement)	120 Marks	Formula Engine: <code>enrolment_ratio</code> , <code>api_year1..3</code> , <code>placement_index</code>
Criterion 5	Faculty Information (SFR, Cadre Ratio, FQI, Retention)	100 Marks	Formula Engine: <code>student_faculty_ratio</code> , <code>faculty_qualification_index</code>
Criterion 6	Faculty Contributions (FDPs, Research Papers, Patents, Grants)	120 Marks	Formula Engine: <code>research_funding_score</code> , <code>consultancy_score</code>
Criterion 7	Facilities & Technical Support (Laboratories, Ambiance, Safety)	100 Marks	Infrastructure Registry + Lab Inventory
Criterion 8	Continuous Improvement (Academic Audit, Action Taken on Attainment)	80 Marks	OBE Actions + Longitudinal Comparative Trends
Criterion 9	Student Support & Governance (Mentoring, Budget, Feedback, FPADS)	120 Marks	Formula Engine: <code>first_year_sfr</code> + Governance Records
TOTAL NBA TIER-II EVALUATION SCORE		1000 Marks	Exact Leaf Marks Sum Verified via Assertions

Core Mathematical Indices Implemented in Formula Engine

Student-Faculty Ratio (SFR) — 30 Marks

$$\text{SFR} = \text{Total Approved Intake} / \text{Total Core Faculty (F)}$$

- Tier-II Marks: $\text{SFR} \leq 15$: 30 marks; $15 < \text{SFR} \leq 20$: Scaled marks; $\text{SFR} > 25$: 0 marks.

Faculty Qualification Index (FQI) — 25 Marks

$$\text{FQI} = 2.5 \times (10X + 6Y + 4Z) / F$$

- X = Ph.D., Y = M.Tech/M.E., Z = B.Tech/Others. Scaled to maximum 25 marks.

Academic Performance Index (API) — 30 Marks

$$\text{API} = \text{Mean CGPA or \% Marks} \div 10 \times (\text{Appeared} / \text{Successful})$$

- Evaluated independently across 1st Year (10M), 2nd Year (10M), and 3rd Year (10M).

Placement & Higher Studies Index — 30 Marks

$$\text{Placement Index} = (\text{Placed} + \text{Higher Studies} + \text{Entrepreneurs}) / \text{Intake}$$

- Score = $30 \times \text{Placement Index}$ (averaged over three assessment years).

4. AI, Machine Learning & Privacy Technologies

A. Intelligent Document OCR & RAG Knowledge Engine

AcademiQ transforms unstructured departmental files (lesson plans, minutes, FDP certificates, inspection guidelines) into a searchable vector knowledge base.

- 1 **Multi-Source Text Ingestion:** Ingestion pipeline processes PDFs, DOCX, and images. Digital PDFs are parsed with PyMuPDF; scanned legacy documents automatically trigger PaddleOCR.
- 2 **Token Chunking:** Documents are partitioned into 512-token chunks with 50-token sliding overlap to preserve boundary context.
- 3 **Dense Embeddings:** The state-of-the-art `BAAI/bge-m3` model embeds chunks into 1024-dimensional vectors stored inside Qdrant.
- 4 **Hallucination-Restricted Generation:** User queries perform cosine-similarity vector retrieval; the top-K chunks are fed to Llama 3.1 8B with a strict system prompt prohibiting ungrounded assertions.

B. Predictive Student Risk Modeling (Dropout Intervention)

The system actively monitors academic risk indicators to allow pro-active faculty intervention before examinations.

8 Predictive Feature Vector

1. `semester` (1 to 8)
2. `attendance_pct` (0–100%)
3. `internal_marks` (0–100%)
4. `assignment_score_pct` (0–100%)
5. `previous_gpa` (0.0–10.0)
6. `backlogs` (count of active backlogs)
7. `course_performance_pct` (0–100%)
8. `engagement_encoded` (Low=0, Med=1, High=2)

Dual Model Pipeline & Retraining

- **Random Forest:** 200 trees, `class_weight="balanced"`, standard scaling pipeline.
- **XGBoost Classifier:** Gradient boosted decision trees for comparative validation.
- **Risk Classification:** High (>0.7), Medium (0.4–0.7), Low (<0.4).
- **Rule Flags:** Auto-flags attendance < 75%, internal marks < 50%, backlogs ≥ 1.

C. Privacy-Preserving Parent Communication (DPDP Act 2023)

DPDP Act Compliance & Masked Telephony: To protect student and guardian privacy under the Digital Personal Data Protection (DPDP) Act:

- **Identity Masking:** Teachers only see masked phone numbers (e.g., `*****3210`) on screen.
- **Twilio Proxy Bridge:** Initiating a call routes through a Twilio cloud proxy. Neither the teacher nor the parent sees the other party's real personal phone number.
- **Explicit Consent Workflow:** Parent communication preferences (Call / SMS / WhatsApp) and consent flags (`consent_to_contact=True`) are verified before outbound requests.

5. Security, Roles & User Interfaces

Role-Based Access Control (RBAC) Matrix

Permissions are strictly enforced at the API Gateway network boundary via a fail-closed routing table:

Platform Feature / Route	Admin	Faculty (Teacher)	Student	Data Worker
Executive Dashboard & Institutional Analytics	Full	Full	Denied	Denied
Student Records & Risk Predictive Profiling	Full	Full	Self Only	Denied
Parent Contact System & Masked Calling	Full	Full	Denied	Denied
Document Upload, OCR & Digitization	Full	Full	Denied	Upload Only
RAG Intelligent Assistant Chat	Full	Full	Full	Denied
NBA SAR Generation & DOCX/PDF Export	Full	Full	History Only	Denied
ML Model Retraining & User Management	Admin Only	Denied	Denied	Denied

Frontend Pages & User Experience (React 18 + Vite)

1. DashboardPage

High-level institutional overview, department KPIs, student distribution, risk alert counters, and quick actions.

2. StudentsPage

Searchable, filterable student roster with section breakdown, attendance charts, and risk level indicators.

3. StudentProfilePage

Comprehensive student dossier with radar charts, internal marks history, ML predictions, and parent details.

4. FacultyPage

Faculty profiles, designation, qualifications, research publications, awards, and FDP certifications.

5. DocumentsPage

Drag-and-drop document uploader, OCR processing queue, classification tags, and status monitoring.

6. RAGChatPage

Conversational AI interface with document citations, question suggestions, and context transparency.

7. ContactPage

Guardian contact interface with masked numbers, consent checks, call triggering, and SMS history.

8. ReportsPage

NBA SAR report generator (full/criterion/subcriterion scope), AI narrative toggles, and direct PDF/DOCX downloads.

9. MyRecordPage

Dedicated student portal showing personal CGPA, attendance alerts, course progress, and academic flags.

6. Key Innovations & Project Impact Summary

Key Project Takeaways:

- **100% Automated NBA SAR Workflow:** Replaces months of manual document collation and spreadsheet formulas with instant, standardized SAR generation compliant with NBA Tier-II GAPC V4.0.
- **Early Student Intervention:** Leverages dual ML models to identify struggling students early in the semester, enabling timely faculty mentoring.
- **Secure Document Intelligence:** Combines local/cloud LLMs with vector search and dual-engine OCR to unlock institutional memory from siloed PDFs.
- **Privacy by Design:** Implements modern DPDP Act compliance through data masking and telephone proxy infrastructure.
- **Enterprise Microservice Engineering:** Demonstrates high modularity, automated seeding, fail-closed gateway security, and Docker container orchestration.

System Deployment & Specifications

Component	Specification / Recommendation
Container Orchestration	<code>Docker Compose v2</code> (11 containers across bridge network <code>academiq-net</code>)
Memory Footprint	8 GB RAM (Groq cloud LLM) or 16 GB RAM (Ollama local Llama 3.1 8B)
Storage Footprint	~5 GB Docker images + 580 MB BGE-M3 model weights
Quick Start Command	<code>docker-compose up --build</code> (One-click launch with auto-seeded demo data)

AcademiQ Platform — Developed by B.E. CSE Z10 Batch (2025–2026)
Final Year Project Demonstration & Architecture Guide